

Regions of Interest (ROIs) and Their Functions

1. Social Anxiety Core Circuit (Threat & Emotion Regulation)

Brain Region	Function
amygdala	Threat detection and fear processing
insula	Interoceptive awareness (e.g., heartbeat, anxiety sensations)
anterior cingulate cortex	Conflict monitoring and emotion regulation
medial orbitofrontal cortex	Reward/punishment evaluation and social valuation
middle frontal gyrus	Cognitive control and top-down regulation
superior frontal gyrus	Attention and executive control

2. Default Mode Network (DMN) - Rumination / Self-referential Processing

Brain Region	Function
posterior cingulate cortex	Self-referential processing and autobiographical reflection
precuneus	Self-consciousness and mental imagery
medial superior frontal gyrus	Self-evaluation and social cognition
anterior cingulate cortex (overlap)	Internal monitoring and introspection

3. Sleep-Arousal Regulation System

Brain Region	Function
hypothalamus	Circadian rhythm regulation (including SCN)
locus coeruleus	Arousal, vigilance, and noradrenergic modulation
dorsal raphe nucleus	Serotonergic regulation of mood and sleep (REM/NREM)
median raphe nucleus	Serotonergic modulation and sleep stabilization
thalamus	Sensory gating and thalamocortical regulation during sleep

4. Reward and Motivation System (Social Avoidance)

Brain Region	Function
nucleus accumbens	Reward processing and hedonic experience
ventral tegmental area	Dopamine production and reward signaling
caudate	Goal-directed learning and habit formation
putamen	Habitual behavior and motor control
pallidum	Output of reward-related motor and motivational signals

Note. Anatomical region nomenclature follows the Automated Anatomical Labeling Atlas 3 (AAL3; Rolls et al., 2020). General functional annotations of all brain regions are based on Kandel et al. (2021). Region-specific functions are further supported by meta-analytic and systematic reviews and large-scale network frameworks. See References.

References

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